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FULL STACK APPLICATION DEVELOPMENT (24SDCS02)

2025-26, Even Semester

Project Problem Statement Submission Form

Sec No-10	Team No-4	Project Title-Skill swap-A Role based Service Exchange system

1.Abstract

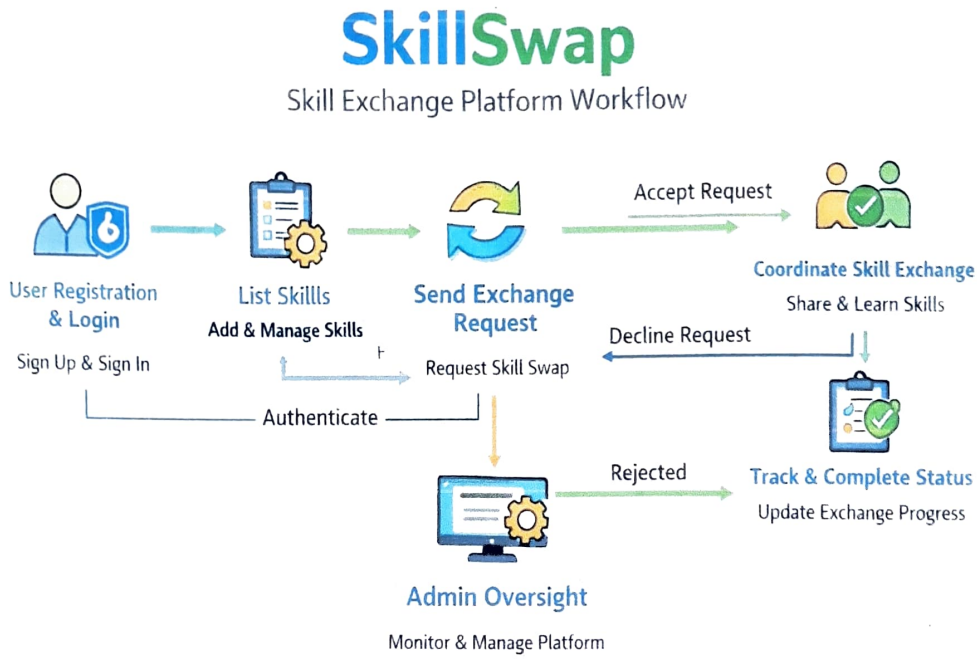
This project presents the design and development of **SkillSwap**, a peer-to-peer skill exchange platform built using Spring Boot and MySQL. The platform addresses a common challenge faced by students who want to learn new skills but have limited access to affordable learning resources, while many students possess valuable skills that can benefit others. Most existing learning platforms focus primarily on paid courses and lack a structured environment for mutual skill sharing.

SkillSwap provides a secure web-based platform where users can register and authenticate themselves through a login and signup system. Authenticated users can list the skills they offer, explore skills shared by others, and send exchange requests based on mutual interests. Once a request is accepted, users coordinate externally to exchange knowledge, while the system tracks the progress of each exchange.

The application implements role-based access control and supports full CRUD operations for users, skills, and exchange requests. A relational database ensures efficient storage, retrieval, and integrity of data. Additionally, an admin module allows monitoring of users and management of skill listings.

This project demonstrates key concepts of Full Stack Application Development, including authentication, database integration, backend processing, and user interface design, making SkillSwap a practical and scalable solution for collaborative learning.

This project demonstrates core concepts of Full Stack Application Development, including authentication, database integration, backend processing, and user interface design, making it a practical and scalable solution for peer-based learning.



2.Module Mapping

Module Name	Assigned Roll Number(s)	Signature
User Authentication & Security	2420030414	GURRAM BHASKER JAI RAJ
Skill Management System	2420090094	MARISA ADITYA VARDHAN
Skill Exchange Request Module	2420030243	MADUPATHY HARI HARAN
Admin Dashboard	2420080035	BOREDDY JAYASIMHA REDDY

3.Module Details

User Authentication & Security:
Responsible for implementing user registration, login functionality, password encryption, and role-based access control to ensure secure platform usage.
Skill Management System:
Handles the creation, updating, deletion, and viewing of skills offered by users. Ensures proper categorization and easy search functionality.
Skill Exchange Request Module:
Manages sending, accepting, rejecting, and tracking exchange requests between users while maintaining the status of each interaction.
Admin Dashboard:
Provides administrative control to monitor users, manage skill listings, and ensure

appropriate platform usage.

4.Frameworks / Platforms / Software Used

Spring Boot:
Spring Boot is used as the primary backend framework to develop the application. It simplifies the configuration of the Spring ecosystem and enables rapid development of production-ready applications. It supports RESTful API creation, dependency management, and embedded servers, allowing the platform to run efficiently without complex setup.

MySQL:
MySQL is used as the relational database management system to store user data, skill listings, and exchange requests. It ensures structured data storage, supports complex queries, maintains data integrity, and provides high reliability for transactional operations.

Spring Data JPA (Hibernate):
Spring Data JPA, implemented with Hibernate, is used for Object Relational Mapping (ORM). It bridges the gap between Java objects and database tables, reducing the need for manual SQL queries and enabling efficient database interaction.

REST APIs:
RESTful web services are implemented to enable communication between the frontend and backend. They allow smooth data transfer, scalability, and support future integration with mobile applications or third-party services.

Frontend Technologies (HTML, CSS, JavaScript):
These technologies are used to design and structure the user interface. HTML provides the layout, CSS enhances visual styling, and JavaScript adds interactivity to improve user experience.

Bootstrap:
Bootstrap is utilized to develop a responsive and mobile-friendly interface. It provides pre-designed components such as forms, navigation bars, and cards, ensuring consistency in design while reducing development time.

IDE (IntelliJ IDEA / Eclipse / VS Code):
An Integrated Development Environment is used for writing, debugging, and testing the code efficiently. It improves productivity through features like code suggestions and error detection.

Git & GitHub:
Git is used for version control to track changes in the project, while GitHub enables collaborative development, code sharing, and backup of the project repository.

Frameworks / Platforms / Software Used

Spring Boot
Backend framework for building and running RESTful APIs, handles dependency management and embedded servers.

MySQL
Relational database for storing user data, skill listings, and exchange requests.

Spring Data JPA (Hibernate)
ORM framework for database access using JPA and Hibernate.

REST APIs
RESTful web services for frontend-backend communication.

Frontend Technologies
HTML, CSS, and JavaScript for UI development.

Bootstrap
Responsive design framework for creating mobile-friendly UI.

IDE (Intellij / Eclipse / VS Code)
Development environments for coding and debugging.

Git & GitHub
Version control and collaboration platform for code management.

Guide Name and Signature (remarks if any) Dr. S. Spandana